Equilibrium

Exam code: 0620 & 0971

Download PDF

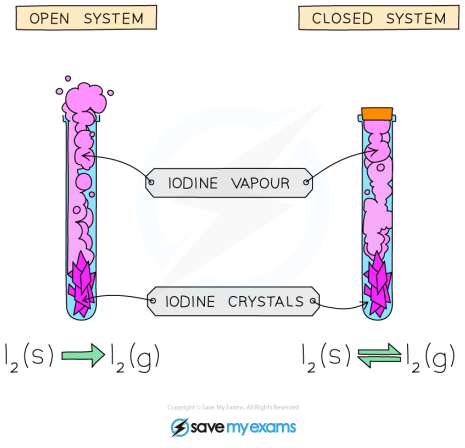
Written by: Alexandra Brennan
Reviewed by: Stewart Hird
Updated on 29 September 2024

Did this video help you? Yes No

The concept of equilibrium

Extended tier only

- Reversible reactions occur in **both** the forward and backward directions
- A reversible reaction can reach **equilibrium** in a **closed system**
  - This is so none of the participating chemical species can leave the reaction vessel and nothing else can enter

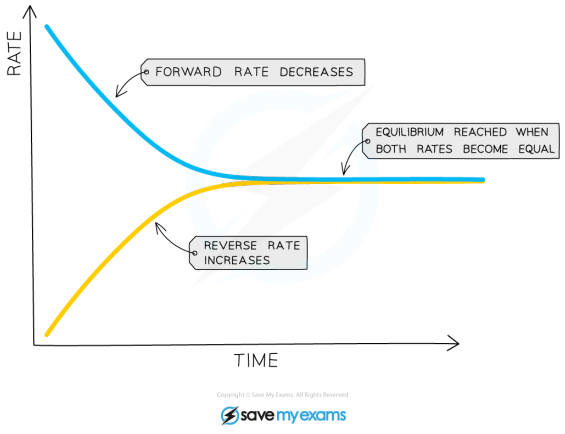


Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- At equilibrium:
  - The rate of the forward reaction is **equal** to the rate of the reverse reaction
  - The **concentration** of reactants and products remains **constant** (given there is no other change to the system such as temperature and pressure)

- Equilibrium is **dynamic**
  - This means that the molecules on the left and right of the equation are **changing** into each other by chemical reactions constantly and at the same rate

- An example of a reaction reaching equilibrium is the reaction between H_2 and N_2 in the Haber process:
  - At the start of the reaction, only nitrogen and hydrogen are present
    - This means that the rate of the forward reaction is at its **highest**, since the **concentrations** of hydrogen and nitrogen are at their **highest**
  - As the reaction proceeds, the **concentrations** of hydrogen and nitrogen gradually **decrease**
    - So, the rate of the forward reaction will **decrease**
  - However, the concentration of ammonia is gradually increasing and so the rate of the **backward** reaction will increase
    - Ammonia will decompose to reform hydrogen and nitrogen
  - In a closed system, the two reactions are interlinked and none of the gases can escape
  - So, the rate of the forward reaction and the rate of the backward reaction will eventually become **equal** and equilibrium is reached:



At equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction

Help me understand

SaveMyExams

Did this video help you? Yes No

Le Chatelier's principle

Extended tier only

- The relative amounts of all the reactants and products at equilibrium depend on the **conditions** of the reaction
  - This balance is framed in an important concept known as Le Chatelier's Principle,
- This principle states that when a change is made to the conditions of a system at equilibrium, the system automatically moves to **oppose** the change
- The principle is used to predict changes to the position of equilibrium when there are changes in:
  - **Temperature**
  - **Pressure**
  - **Concentration**
- Knowing the energy changes, states and concentrations involved allows us to use the principle to manipulate the outcome of reversible reactions
- For example, if pressure is increased, the position of equilibrium moves in the direction which has the smallest amount of gaseous molecules
- The position of equilibrium is said to shift to the **right** when the **forward** reaction is favoured
  - This means that there is an increase in the amount of **products** formed
- The position of equilibrium is said to shift to the **left** when the **reverse** reaction is favoured
  - So, there is an increase in the amount of **reactants** formed

How temperature affects equilibrium

- We can predict the effect of changes in temperature on systems in equilibrium
  - To make this prediction it is necessary to know whether the reaction is exothermic or endothermic
- If the temperature is **raised**:
  - The yield from the endothermic reaction increases
  - The yield from the exothermic reaction decreases

- When a change in temperature is made to a system, the system will **oppose** the change
  - E.g. If the temperature is increased, the system will oppose the change by decreasing the temperature
  - It will do this by favouring the endothermic reaction

The effects of temperature on equilibrium

Change	How the equilibrium shifts
Increase in temperature	Equilibrium moves in the **endothermic** direction to reverse the change
Decrease in temperature	Equilibrium moves in the **exothermic** direction to reverse the change

Worked Example

Iodine monochloride reacts reversibly with chlorine to form iodine trichloride.

The forward reaction is exothermic:

$$ICl + Cl_2 \rightleftharpoons ICl_3$$

dark brown yellow

What colour will the mixture turn when heated? Explain your answer.

Answer:

- The system will oppose the increase in temperature
  - Increasing the temperature of an equilibrium reaction favours the endothermic reaction
- If the forward reaction is exothermic, then the backward reaction must be endothermic
- Therefore, the equilibrium will move to the left and produce more of the reactants
- This means that the colour of the mixture will become increasingly brown as the temperature increases

How pressure affects equilibrium

- Changes in pressure only affect reactions where the reactants or products are gases
- We can predict the effect of changes in pressure on systems in equilibrium
  - To make this prediction, the balanced symbol equation must be known

The effects of pressure on equilibrium

Increase in pressure	Equilibrium shifts in the direction that produces the **least** number of molecules
Decrease in pressure	Equilibrium shifts in the direction that produces the **greatest** number of molecules

Worked Example

Nitrogen dioxide molecules can dimerise and form dinitrogen tetroxide in the following equilibrium reaction:

$$\underset{\text{brown gas}}{2NO_2(g)} \rightleftharpoons \underset{\text{colourless gas}}{N_2O_4(g)}$$

What will the colour change be if the pressure is increased? Explain your answer.

Answer:

- The number of gas molecules produced by the forward reaction = 1
- The number of gas molecules produced by the reverse reaction = 2
- An **increase** in pressure will cause equilibrium to shift in the direction that produces the **least** number of molecules of gas
  - This is the forward reaction
- So, the equilibrium shifts to the right
- This means that:
  - The mixture will become increasingly colourless
  - The concentration of N_2O_4 will increase

How concentration affects equilibrium

- The effect of changing concentration can be thought of as a balance, with the reactants on the left and the products on the right
  - If the concentration of a reactant increases, then the equilbrium shifts to the right to balance this balance

The effects of concentration on equilibrium

Change	How the equilibrium shifts
Increase in concentration of a reactant	Equilibrium shifts to the right

Increase in concentration of a product	Equilibrium shifts to the left
Decrease in concentration of a product	Equilibrium shifts to the right

Worked Example

Iodine monochloride reacts reversibly with chlorine to form iodine trichloride

$$ICl + Cl_2 \rightleftharpoons ICl_3$$

dark brown yellow

Explain what happens when:

1. The concentration of ICl_3 increases
2. Some Cl_2 is removed

Answers:

1. The concentration of ICl_3 increases
   - There are more molecules of ICl_3 on the products side
   - So, the position of the equilibrium moves **to the left**
   - This produces more ICl and Cl_2
     - So, the reaction mixture gets darker / turns dark brown
2. Some Cl_2 is removed
   - There are less molecules of Cl_2 on the reactants side
   - So, the position of the equilibrium moves **to the left**
   - This produces more Cl_2 (and ICl)
     - So, the reaction mixture gets darker / turns dark brown

How catalysts affects equilibrium

- The presence of a catalyst:
  - Does **not** affect the position of equilibrium
  - Increases the rate at which equilibrium is reached
- This is because the catalyst increases the rate of **both** the forward and backward reactions by the same amount by providing an alternative pathway requiring lower activation energy
- As a result, the equilibrium **concentration** of reactants and products is the **same** as it would be without the catalyst

Diagram showing the effect of a catalyst on the time taken for equilibrium to be established

Examiner Tips and Tricks

When the conditions at equilibrium are changed, the system always responds by doing the **opposite**.

For example if the concentration is increased the system tries to reduce it by changing the direction of the reaction or if the temperature is increased the system will try to reduce the temperature by absorbing the extra heat.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.